

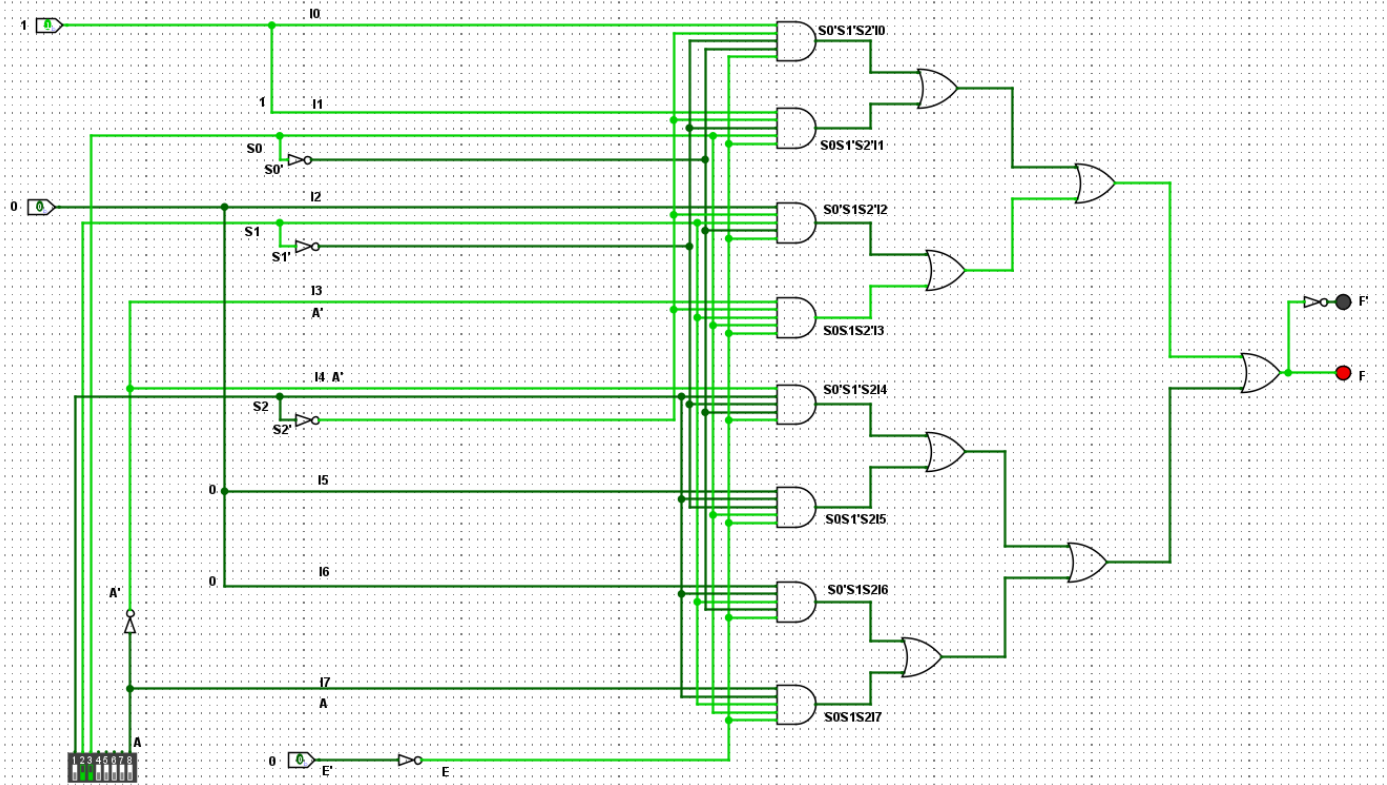


Figure 1; Circuit Diagram using logic gates

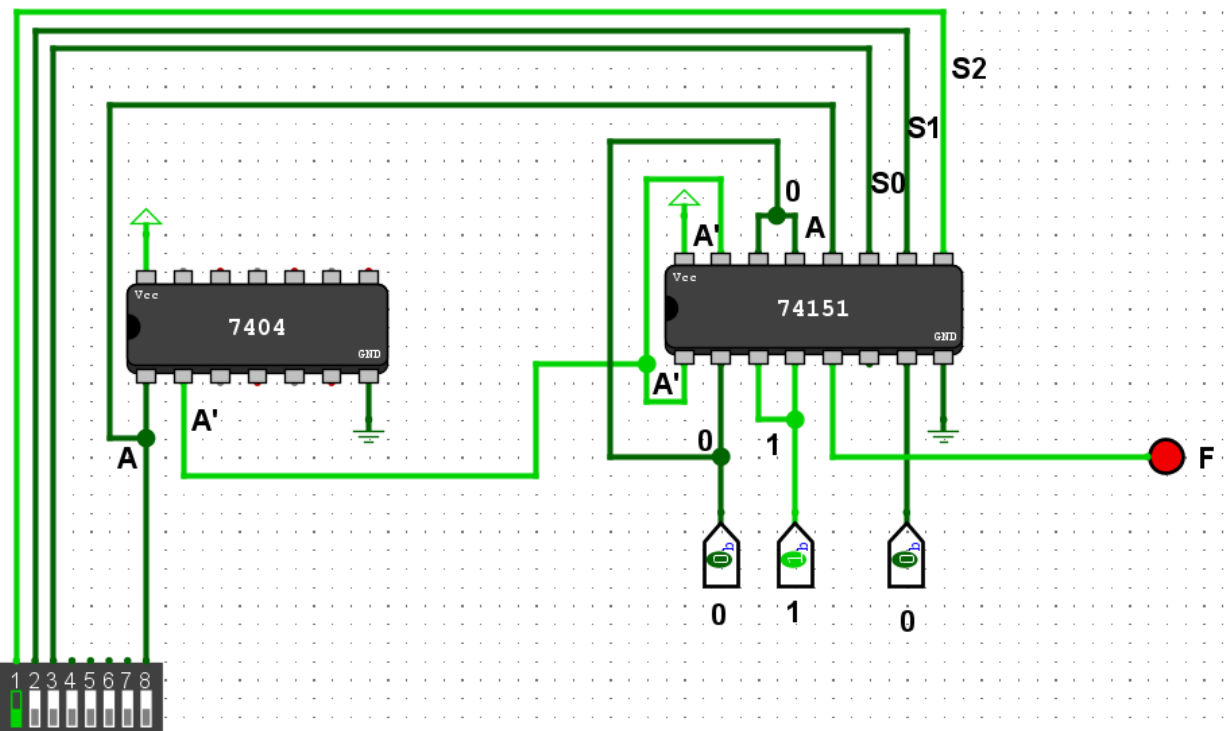


Figure 2: Circuit Diagram using IC's

```

module mux (
    input wire [2:0] S ,
    input wire A,
    output reg Y
);

wire [7:0] I;

assign I[0]= 1'b1;
assign I[1]= 1'b1;
assign I[2]= 1'b0;
assign I[3]= ~A;
assign I[4]= ~A;
assign I[5]= 1'b0;
assign I[6]= 1'b0;
assign I[7]= A;

always @(*) begin
    case (S)
        3'b000: Y = I[0];
        3'b001: Y = I[1];
        3'b010: Y = I[2];
        3'b011: Y = I[3];
        3'b100: Y = I[4];
        3'b101: Y = I[5];
        3'b110: Y = I[6];
        3'b111: Y = I[7];
        default: Y = 1'b0;
    endcase
end

endmodule

```

Truth Table for A = 0						
S2	S1	S0	Input	F		
0	0	0	I0	1		
0	0	1	I1	1		
0	1	0	I2	0		
0	1	1	I3	1		
1	0	0	I4	1		
1	0	1	I5	0		
1	1	0	I6	0		
1	1	1	I7	0		

Truth Table for A = 1						
S2	S1	S0	Input	F		
0	0	0	I0	1		
0	0	1	I1	1		
0	1	0	I2	0		
0	1	1	I3	0		
1	0	0	I4	0		
1	0	1	I5	0		
1	1	0	I6	0		
1	1	1	I7	1		

```

`timescale 1ns / 1ps

module mux_tb;
    reg [2:0] S;
    reg A;
    wire Y;
    integer i;

    mux uut(
        .S(S),
        .A(A),
        .Y(Y)
    );

    reg clk;
    initial clk = 0;
    always #5 clk = ~clk;

    initial begin
        $dumpfile("waveform.vcd");
        $dumpvars(0, mux_tb);

        // Test for A = 0
        $display("\n\n\n=====");
        $display("    Truth Table for A = 0 ");
        $display("=====");
        $display(" S2  S1  S0 | Input | F |");
        $display("-----");

        A = 0;

        for (i = 0; i < 8; i++) begin
            S = i;
            #10;
            $display(" %b  %b  %b | I%-2d  | %b |", S[2], S[1], S[0], i, Y);
        end
    end
endmodule

```

```

41 // Test for A = 1
42 $display("\n\n\n=====");
43 $display("    Truth Table for A = 1 ");
44 $display("=====");
45 $display(" S2  S1  S0 | Input | F |");
46 $display("-----");
47
48 A = 1;
49
50 for (i = 0; i < 8; i++) begin
51     S = i;
52     #10;
53     $display(" %b  %b  %b | I%-2d  | %b |", S[2], S[1], S[0], i, Y);
54 end
55
56 repeat (20) begin
57     @(posedge clk);
58
59     S[2] = $random %2;
60     S[1] = $random %2;
61     S[0] = $random %2;
62     A = $random %2;
63     #1;
64
65     $display(" %b  %b  %b | %b | %b", S[2], S[1], S[0], A, Y);
66
67 end
68
69 $display("\nSimulation Finished.");
70 $finish;
71
72 end
73
74 endmodule
75

```



Figure 3: Verilog Code with corresponding Truth table and Signal Response for 8 to 1 line Multiplexer